CNSH: A Human-Centered AI Governance and Knowledge Architecture

With CNSH-64 Decision-State Model, CNSH-Lang Governance

Automation Language, and CNSH-OS Seven-Layer Blueprint

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Abstract—This paper introduces CNSH, a human-centered architecture designed for AI-assisted knowledge systems and governance frameworks. The architecture integrates ethical constraints, decision-state reasoning models, and transparent audit structures to ensure that AI systems remain accountable, interpretable, and aligned with human values.

CNSH proposes a three-layer governance structure consisting of a Principle Layer (governance logic), a System Layer (data and automation), and an Interaction Layer (human interface). The framework combines symbolic decision-state modeling (the CNSH-64 model with 8 base states composing 64 decision states), structured knowledge management via knowledge cards and knowledge graphs, and automated auditing mechanisms.

We further present CNSH-Lang, a governance automation language with near-natural-language syntax and built-in audit mechanisms; CNSH-OS, a seven-layer operating system blueprint for human-centered AI collaboration; and the CNSH System Constitution, a seven-article constitutional framework establishing the foundational principles of transparent, accountable AI governance.

The goal of CNSH is to create a transparent and resilient AI collaboration environment, enabling humans and intelligent systems to operate within clear ethical and structural boundaries.

Index Terms—CNSH, AI governance, human-centered architecture, decision-state model, knowledge graph, ethical constraint engine, audit system, governance automation language, CNSH-Lang, CNSH-OS, system constitution

I. Introduction

Modern AI systems are rapidly expanding in capability but often lack transparent governance structures. Key challenges include:

- lack of traceable decision mechanisms,
- absence of ethical boundaries,
- fragmented knowledge systems,
- insufficient accountability in automated systems.

CNSH addresses these problems by introducing a unified architecture that integrates ethical governance, symbolic reasoning structures, transparent data systems, and human-centered interaction design. The system does not attempt to replace human decision-making. Instead, it focuses on augmenting human reasoning through structured AI assistance.

TABLE I
CNSH Three-Layer Governance Architecture

Layer	Function
Principle Layer	Governance logic and decision models
System Layer	Structured data and automation
Interaction Layer	Human interface and operational tools

A. Core Design Philosophy

The CNSH framework follows five core design principles:

1) **Transparency**: All decisions and system actions must be traceable.
2) **Responsibility**: Every automated action must have a verifiable origin.
3) **Ethical Alignment**: System operations must remain within defined ethical boundaries.
4) **Human Priority**: Technology serves people rather than replacing human judgment.
5) **Knowledge Continuity**: Knowledge must remain understandable and maintainable over long time periods.

B. The Three-Layer Governance Architecture

CNSH implements a hierarchical system architecture as shown in Table I.

This separation improves maintainability, interpretability, and system scalability.

II. The CNSH-64 Decision-State Reasoning Model

CNSH introduces a symbolic decision-state model to guide AI-supported reasoning. The model represents operational states of a system through 8 base states that combine to form 64 decision states.

A. Base States

B. State Composition

States combine to form complex reasoning contexts. For example:

TABLE II
CNSH-64 Base States

State ID	Symbol	Meaning
S1	Initiation	System creation or new action
S2	Foundation	Structural foundation
S3	Trigger	Event activation
S4	Propagation	Communication or influence
S5	Risk	Uncertainty detection
S6	Awareness	Observation and analysis
S7	Boundary	Limitation control
S8	Cooperation	Coordinative interaction

TABLE III
Decision Classification Levels

Level	Result
□ Green	Safe to execute
□ Yellow	Conditional execution
□ Red	Blocked or restricted

- **Foundation + Trigger**: Stable system encountering change,
- **Risk + Boundary**: Potential threat under constraint,
- **Initiation + Cooperation**: New collaborative action.

These states function as reasoning primitives, allowing complex situations to be interpreted through structured symbolic states.

C. Decision Flow

The CNSH-64 decision flow follows a six-stage pipeline:

1) **Input Event**: Receive and parse external event,
2) **State Identification**: Map event to base state(s),
3) **State Combination**: Compose 64-state model,
4) **Governance Evaluation**: Apply governance rules,
5) **Ethical Constraint Check**: Validate ethical compliance,
6) **Action Classification**: Output decision level.

D. Decision Classification

E. Automation Trigger Tags

III. Ethical Constraint Engine

A built-in ethical constraint engine evaluates actions before execution. Evaluation dimensions include:

- social impact,
- operational risk,
- ethical compliance.

Results are categorized using the three-level classification system described in Section III. This model provides a lightweight but effective governance structure for automated systems.

Proposition 1 (Ethical Constraint Completeness). For any system action a, the ethical constraint engine $\mathcal{E}$ produces a classification $\mathcal{E}(a) \in \{\text{GREEN}, \text{YELLOW}, \text{RED}\}$ such that:

$$\mathcal{E}(a) = \text{RED} \implies a \text{ is blocked} \tag{1}$$

TABLE IV
Automation Trigger Tags

Tag	Function
`#decision`	Decision record
`#risk`	Risk identification
`#alert`	System warning
`#governance`	Governance action
`#audit`	Audit record

and:

$$\mathcal{E}(a) = \text{YELLOW} \implies a \text{ requires additional human review.} \tag{2}$$

IV. Knowledge Architecture

CNSH uses a structured knowledge-card system as its fundamental knowledge unit.

A. Knowledge Card Structure

Each knowledge unit includes:

- title,
- summary,
- source,
- content,
- tags,
- relationships.

Knowledge cards form a **knowledge graph**, enabling large-scale information organization and semantic retrieval.

B. Knowledge Graph

The knowledge graph $\mathcal{G} = (V, E, W)$ consists of:

- **Nodes** V: knowledge cards, decisions, events, resources,
- **Edges** E: references, influences, dependencies,
- **Weights** W: semantic proximity scores.

V. Audit and Traceability

All system actions generate audit records. Each record contains:

- event ID,
- action type,
- operator,
- timestamp,
- result.

This ensures transparency, accountability, and forensic analysis capability.

Proposition 2 (Audit Completeness). For every system operation op, there exists a unique audit record r_{op} such that:

$$\forall op \in \mathcal{O} : \exists! r_{op} \in \mathcal{R} \quad \text{where} \quad r_{op} = (\text{id}, \text{type}, \text{operator}, \text{timestamp}, \text{resu} \tag{3}$$

where $\mathcal{O}$ is the set of all operations and $\mathcal{R}$ is the audit record set.

VI. Automation and System Health

Automated monitoring maintains system integrity. Monitoring includes:

- schema consistency,
- data completeness,
- duplicate detection,
- orphan node detection,
- broken relation detection.

When issues are detected, automated repair tasks are generated with priority levels and notification mechanisms.

A. System Health Cycle

The health monitoring loop operates continuously:

1) **Scan**: Inspect all system components,
2) **Detect**: Identify anomalies and issues,
3) **Repair**: Generate and execute repair tasks,
4) **Verify**: Confirm issue resolution.

B. Three Automation Cycles

CNSH implements three continuous automation cycles:

1) Knowledge Cycle:

   input content → clean → summarize → tag → store → connect graph

2) Governance Cycle:

   event → evaluate state → ethical check → decision classification → record audit

3) System Health Cycle:

   scan system → detect anomalies → generate repair tasks → verify fixes

VII. CNSH-Lang: Governance Automation Language

CNSH-Lang is a governance automation language designed for human readability and built-in audit mechanisms. Its core characteristics include:

- near-natural-language syntax,
- built-in governance and audit mechanisms,
- explicit constraint declaration,
- automated task scheduling.

A. Action Declaration

```
ACTION create_knowledge
  INPUT article
  OUTPUT knowledge_card
  TAG #learning
```

B. Decision Rule

```
RULE evaluate_risk
  IF risk_score > 70
    THEN classification = RED
  ELSE classification = GREEN
```

C. Governance Constraint

```
CONSTRAINT ethical_check
  IF action_impact = high
    REQUIRE human_review
```

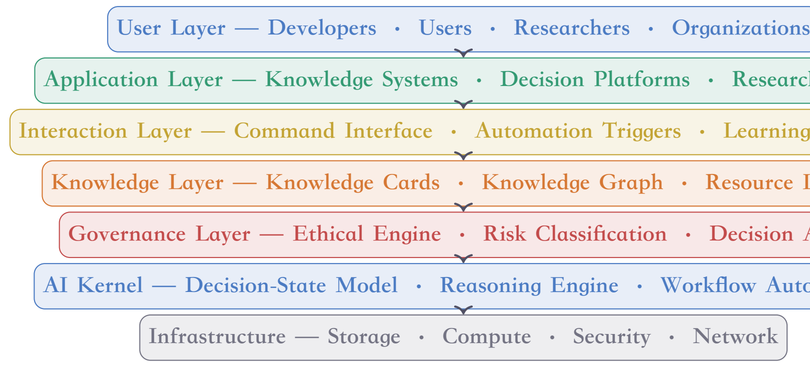

Fig. 1. CNSH-OS seven-layer system architecture. Each layer is independently extensible with clearly defined responsibilities.

D. Knowledge Conversion Pipeline

```
PROCESS learning_pipeline
  INPUT raw_content
  STEP clean_text
  STEP summarize
  STEP generate_card
  STEP tag
  STEP store_database
```

E. Automated Task

```
TASK system_health_scan
  RUN every 24h
  CHECK
    database_integrity
    orphan_nodes
    missing_properties
    duplicate_cards
```

F. Audit Record Format

```
AUDIT_LOG
  event_id
  operator
  timestamp
  action_type
  result
```

VIII. CNSH-OS: Seven-Layer Operating System Blueprint

CNSH-OS is a human-centered AI operating system architecture designed for the AI collaboration era. Its core goal is not to replace traditional systems, but to provide a unified runtime environment for AI governance, knowledge management, and automated collaboration.

A. Seven-Layer Architecture

B. AI Kernel

The AI Kernel is the core engine of CNSH-OS, containing four critical sub-modules:

- **State Model**: The CNSH-64 decision-state model,
- **Reasoning Engine**: Symbolic reasoning over combined states,

- **Workflow Automation**: Event-driven workflow scheduling,
- **Learning Processor**: Knowledge extraction and structuring.

C. Governance Kernel

The Governance Kernel distinguishes CNSH-OS from traditional AI systems. Its core responsibilities include: AI behavior constraint, decision transparency, and risk control.

The governance pipeline:

> action request → risk evaluation → ethical validation → decision classification → execution or block → audit log

D. Knowledge Kernel

The knowledge layer is the source of long-term system value. Knowledge units are organized as Knowledge Cards with fields: summary, content, source, tags, and relations. Cards form a knowledge graph for semantic navigation.

E. Identity Kernel

The identity system ensures:

- unique user identification,
- clear permission levels,
- auditable contribution records.

Identity fields include: `identity_id`, `dna_hash`, `public_key`, `access_level`, `contribution_score`.

F. Automation Engine

> event trigger → state evaluation → workflow selection → execution → audit recording

Automation modules: workflow scheduler, trigger system, repair engine, monitoring engine.

G. Security Architecture

- identity verification,
- access control,
- audit monitoring,
- policy enforcement.

H. Application Scenarios

CNSH-OS supports diverse domains:

- AI governance systems,
- research knowledge management,
- enterprise decision platforms,
- education systems,
- open-source communities.

I. Future Extensions

- distributed knowledge networks,
- local AI model integration,
- open-source ecosystem,
- cross-organization collaboration systems.

IX. CNSH System Constitution

The CNSH Constitution defines the foundational principles, governance mechanisms, and technical boundaries of the system. Its goal is to ensure that technological development always serves human society and remains transparent, fair, and accountable.

A. Article 1 — Human Priority

The primary goal of technology is to enhance human capability. No automated system may diminish human decision-making authority or agency. AI provides assistance; humans retain final judgment. Important decisions must allow human intervention.

B. Article 2 — Transparency

System behavior must remain interpretable. Every system action must be traceable to: behavior origin, execution logic, result record. Transparency is achieved through the Audit System.

C. Article 3 — Accountability

All system actions must produce verifiable records including: operating subject, action type, execution time, execution result. These records constitute the system's accountability mechanism.

D. Article 4 — Ethical Boundary

Technical behavior must be constrained by ethics. The Ethical Engine evaluates actions on: risk level, social impact, ethical compliance. High-risk actions require human review.

E. Article 5 — Knowledge Continuity

The system must protect and perpetuate knowledge. Knowledge structures must have: long-term readability, traceable sources, clear structure. Knowledge is stored as Knowledge Cards forming a knowledge network.

F. Article 6 — Responsible Automation

Automation must be governed:

> event trigger → governance evaluation → ethical validation → execution → audit record

G. Article 7 — System Integrity

The system must possess self-detection capability. Regular checks include: data integrity, system structure, data relationships. Issues automatically generate repair tasks.

X. AI Governance Standard Proposal

This section proposes a basic framework for AI system governance, structured similarly to ISO/IEEE standards.

A. Governance Framework

AI systems should include four governance components:

B. Decision Governance Model

> input → context analysis → risk evaluation → ethical validation → decision classification → execution

TABLE V
Required Governance Components

Component	Function
Audit System	Behavior tracking
Human Oversight	Human supervision
Ethical Engine	Constraint validation
Decision Model	State reasoning

TABLE VI
AI Risk Classification Standard

Level	Handling
□ Low	Normal execution
□ Medium	Monitored execution
□ High	Human review required

C. Risk Classification

D. AI Accountability

AI systems must possess:

- behavior records,
- explanation capability,
- traceable decisions.

E. Human Oversight

Any automated system must allow:

- pause execution,
- modify decisions,
- force termination.

F. Governance Audit

Periodic governance audit checks:

- AI behavior records,
- decision quality,
- risk management.

XI. CNSH Core Principle & Technology Declaration

A. Core Declaration

> **Technology should enhance human capability while remaining transparent, accountable, and ethically bounded.**

B. Principle-to-System Mapping

C. System Enforcement Pipeline

> User Request → AI Assistance → Ethical Validation → Execution → Audit Record

D. Engineering Rules

1) AI does not directly replace human decision-making.
2) All system behavior must be recorded.
3) Any high-risk action must pass through the governance module.
4) Automation must allow human intervention.

TABLE VII
Principle-to-System Module Mapping

Principle	System Module
Transparency	Audit System
Accountability	DNA Traceability
Ethical Boundary	Ethical Engine
Human Priority	Human Override Interface

E. Governance Control

> action → risk evaluation → ethical validation → approval or block

F. Transparency Mechanism

All behavior is written to the Audit Log with fields: action, operator, timestamp, result.

G. Human Override

Three override levels:

- **pause**: Temporarily halt execution,
- **modify**: Alter decision parameters,
- **cancel**: Force termination.

H. Extended Governance Principles

- **Knowledge Preservation**: Knowledge must be accumulable and traceable.
- **Responsible Automation**: Automation must be governed.
- **Human Sovereignty**: Final decision authority belongs to humans.

I. Technology Declaration

> Technology must serve humanity. Artificial intelligence should augment human capability rather than replace human judgment. Systems must remain transparent, accountable, and governed by ethical boundaries. Innovation without responsibility leads to instability. Responsible technology creates sustainable progress.

J. Core Beliefs

- Human values first,
- Technology must be governable,
- Knowledge must be openly accumulated,
- Innovation must bear responsibility.

K. Vision

CNSH's goal is to establish a new technological environment for human-intelligent-system collaboration, in which: humans maintain agency, AI provides capability extension, systems remain transparent, and knowledge continuously accumulates.

L. Core Formula

$$\text{AI Capability} + \text{Human Judgment} + \text{Ethical Governance} = \text{Responsible In} \tag{4}$$

M. System Motto

Human first. Technology accountable. Intelligence governed.

XII. Applications

CNSH can support various domains:

- **Research knowledge systems:** Structured knowledge management with semantic retrieval,
- **AI governance platforms:** Ethical constraint enforcement and decision auditing,
- **Enterprise decision systems:** Transparent decision-state tracking,
- **Education platforms:** Knowledge card-based learning with graph navigation,
- **Open knowledge communities:** Collaborative knowledge accumulation with provenance tracking.

XIII. Conclusion

CNSH proposes a human-centered architecture for AI governance and knowledge management.

By combining symbolic reasoning models (CNSH-64), ethical constraints, transparent audit systems, and a governance automation language (CNSH-Lang), CNSH aims to create AI systems that remain accountable, interpretable, and aligned with human values.

The seven-layer CNSH-OS blueprint provides a practical engineering framework, while the CNSH System Constitution establishes foundational principles that ensure technology always serves humanity.

The core formula (Eq. 4) captures the essential insight: responsible intelligence emerges from the combination of AI capability, human judgment, and ethical governance—not from any single component alone.

Future work will explore distributed deployment, integration with local AI models, and open-source collaboration.

> Human first. Technology accountable. Intelligence governed.

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The CNSH architecture builds upon Chinese philosophical traditions including the I Ching (Book of Changes), Tao Te Ching, and the Three-Powers (San-Cai) framework of Heaven – Earth – Human unity, integrating them with modern AI governance requirements.

Appendix

```
      USER LAYER
developers · users · researchers · orgs

      APPLICATION LAYER
knowledge systems · decision platforms
research tools · project management

      INTERACTION LAYER
command interface · automation triggers
learning interface · monitoring

      KNOWLEDGE LAYER
knowledge cards · knowledge graph
resource library · structured databases

      GOVERNANCE LAYER
ethical engine · risk classification
decision audit · policy rules

      AI KERNEL
decision-state model · reasoning engine
workflow automation · learning proc.

      INFRASTRUCTURE
storage · compute · security · network
```

Module	Status	Description
Identity Layer		User identity governance
Governance Layer		Decision & ethics engine
Interaction Layer		Human interface tools
System Layer		Structured data & automation
Automation Engine		Workflow & repair engine
Knowledge Graph		Semantic knowledge network
Resource Library		Papers, datasets, code, docs
Audit System		Action tracking & forensics
Health Monitor		System integrity scanning
AI Integration		Local & external AI models
Infrastructure Layer		Storage, compute, security
CNSH-64 Decision Model		8 base × 64 combined states
CNSH-Lang		Governance automation language
CNSH-OS Blueprint		Seven-layer OS architecture
CNSH Constitution		Seven-article governance charte

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